Test Case Document					
Group Number - 29					Shobikan V. 214197C
Test Identifier	**Test Summery**	**Test Description**	**Precondition**	**Test Steps**	**Expected Result**
API Test Cases – Admin Role					
TC_API_CAT_ADMIN_01	Admin creates main category	Admin creates a main category (no parent)	Admin is logged in	1. Send POST request to /api/categories with valid name and no parentId. 2. Validate response.	1. Status code 201 Created. 2. Response body includes new category details.
TC_API_CAT_ADMIN_02	Admin creates sub-category	Admin creates a sub-category with valid parent	Admin is logged in, parent category exists	1. Send POST request with name and valid parentId. 2. Validate response.	1. Status code 201 Created. 2. Response includes parent category reference.
TC_API_CAT_ADMIN_03	Admin updates category name	Admin updates the name of an existing category	Admin is logged in, category exists	1. Send PUT request to /api/categories/{id} with new name. 2. Validate response.	1. Status code 200 OK. 2. Updated name reflected in response.
TC_API_CAT_ADMIN_04	Admin deletes category	Admin deletes a category without sub-categories	Admin is logged in, category exists	1. Send DELETE request to /api/categories/{id}. 2. Validate response.	1. Status code 204 No Content. 2. Category is removed from system.
TC_API_CAT_ADMIN_05	Admin fetches category by ID	Admin retrieves a single category by ID	Admin is logged in, category exists	1. Send GET request to /api/categories/{id}. 2. Validate response.	1. Status code 200 OK. 2. Correct category details returned.
TC_API_CAT_ADMIN_06	Admin creates duplicate category	Admin attempts to create category with duplicate name under same parent	Admin is logged in, category name exists under same parent	1. Send POST request with duplicate name and same parentId. 2. Validate response.	1. Status code 400 Bad Request. 2. Error message indicates duplicate name.
TC_API_CAT_ADMIN_07	Admin creates category with short name	Admin attempts to create category with name < 3 characters	Admin is logged in	1. Send POST request with name length. 2. Validate response.	1. Status code 400 Bad Request. 2. Validation error for name length.
TC_API_CAT_ADMIN_08	Admin creates category with long name	Admin attempts to create category with name > 10 characters	Admin is logged in	1. Send POST request with name length 2. Validate response.	1. Status code 400 Bad Request. 2. Validation error for name length.
TC_API_CAT_ADMIN_09	Admin deletes category with sub-categories	Admin attempts to delete a category that has child categories	Admin is logged in, category has children	1. Send DELETE request to category with sub-categories. 2. Validate response.	1. Status code 400 Bad Request. 2. Error message indicates category has children.
TC_API_CAT_ADMIN_10	Admin updates category with invalid parent	Admin attempts to assign non-existent parent to a category	Admin is logged in, category exists	1. Send PUT request with non-existent parentId. 2. Validate response.	1. Status code 400 Bad Request. 2. Error message indicates invalid parent.
TC_API_CAT_ADMIN_11	Admin fetches non-existent category	Admin attempts to fetch category by invalid ID	Admin is logged in	1. Send GET request to /api/categories/{invalid-id}. 2. Validate response.	1. Status code 404 Not Found. 2. Appropriate error message.
TC_API_CAT_ADMIN_12	Admin searches categories by name	Admin searches categories using partial name match	Admin is logged in	1. Send GET request to /api/categories?name=partial. 2. Validate response.	1. Status code 200 OK. 2. List includes matching categories.
TC_API_CAT_ADMIN_13	Admin filters categories by parent	Admin filters categories by parentId	Admin is logged in	1. Send GET request to /api/categories?parentId={id}. 2. Validate response.	1. Status code 200 OK. 2. Only sub-categories of given parent returned.

TC_API_CAT_ADMIN_14	Admin paginates categories with sorting	Admin retrieves paginated categories with sorting applied	Admin is logged in	1. Send GET request with page, size, sortBy=name. 2. Validate response.	1. Status code 200 OK. 2. Paginated and sorted list returned.
API Test Cases – User Role					
TC_API_CAT_USER_01	User fetches all categories	User retrieves list of all categories	User is logged in	1. Send GET request to /api/categories. 2. Validate response.	1. Status code 200 OK. 2. List of all categories returned.
TC_API_CAT_USER_02	User fetches category by ID	User retrieves a single category by ID	User is logged in, category exists	1. Send GET request to /api/categories/{id}. 2. Validate response.	1. Status code 200 OK. 2. Category details returned.
TC_API_CAT_USER_03	User searches categories by name	User searches categories using name filter	User is logged in	1. Send GET request with name query parameter. 2. Validate response.	1. Status code 200 OK. 2. Matching categories returned.
TC_API_CAT_USER_04	User fetches sub-categories	User filters categories by parentId	User is logged in	1. Send GET request with parentId parameter. 2. Validate response.	1. Status code 200 OK. 2. Only sub-categories returned.
TC_API_CAT_USER_05	User fetches main categories	User retrieves only main categories (no parent)	User is logged in	1. Send GET request with parentId=null. 2. Validate response.	1. Status code 200 OK. 2. Only main categories returned.
TC_API_CAT_USER_06	User tries to create category	User attempts to create a new category	User is logged in	1. Send POST request to /api/categories with valid data. 2. Validate response.	1. Status code 403 Forbidden. 2. Error message indicates insufficient permissions.
TC_API_CAT_USER_07	User tries to update category	User attempts to update an existing category	User is logged in	1. Send PUT request to /api/categories/{id}. 2. Validate response.	1. Status code 403 Forbidden. 2. Error message indicates insufficient permissions.
TC_API_CAT_USER_08	User tries to delete category	User attempts to delete a category	User is logged in	1. Send DELETE request to /api/categories/{id}. 2. Validate response.	1. Status code 403 Forbidden. 2. Error message indicates insufficient permissions.
TC_API_CAT_USER_09	User fetches non-existent category	User attempts to fetch category by invalid ID	User is logged in	1. Send GET request to /api/categories/{invalid-id}. 2. Validate response.	1. Status code 404 Not Found. 2. Appropriate error message.
TC_API_CAT_USER_10	User paginates categories	User retrieves paginated list of categories	User is logged in	1. Send GET request with page and size parameters. 2. Validate response.	1. Status code 200 OK. 2. Paginated list returned.
UI Test Cases – Admin Role					
TC_UI_CAT_ADMIN_01	Admin sees Category List with pagination	Admin views category list with pagination controls	Admin is logged in	1. Navigate to /ui/categories. 2. Verify list and pagination controls.	1. Category list displayed. 2. Pagination buttons visible if more than one page exists.
TC_UI_CAT_ADMIN_02	Admin sees "Add Category" button	Admin views the Add Category button on list page	Admin is logged in	1. Navigate to /ui/categories. 2. Look for "Add Category" button.	1. "Add Category" button is visible.
TC_UI_CAT_ADMIN_03	Admin can add main category	Admin creates a main category (no parent)	Admin is logged in	1. Click "Add Category". 2. Enter valid name, leave parent empty. 3. Submit.	1. New main category appears in list. 2. No parent category assigned.
TC_UI_CAT_ADMIN_04	Admin can add sub-category	Admin creates a sub-category with parent selection	Admin is logged in, main category exists	1. Click "Add Category". 2. Enter name, select parent. 3. Submit.	1. New sub-category appears under selected parent.

TC_UI_CAT_ADMIN_05	Admin can edit existing category	Admin edits the name of an existing category	Admin is logged in, category exists	1. Click edit icon next to category. 2. Change name, submit.	1. Updated name reflected in list.
TC_UI_CAT_ADMIN_06	Admin can delete category with confirmation	Admin deletes a category with confirmation modal	Admin is logged in, category exists	1. Click delete icon. 2. Confirm deletion in modal.	1. Category removed from list. 2. Confirmation modal disappears.
TC_UI_CAT_ADMIN_07	Admin sees validation error for short name	Admin attempts to add category with name < 3 chars	Admin is logged in	1. Click "Add Category". 2. Enter name < 3 chars. 3. Submit.	1. Red error message appears. 2. Form not submitted.
TC_UI_CAT_ADMIN_08	Admin sees validation error for long name	Admin attempts to add category with name > 10 chars	Admin is logged in	1. Click "Add Category". 2. Enter name > 10 chars. 3. Submit.	1. Red error message appears. 2. Form not submitted.
TC_UI_CAT_ADMIN_09	Admin can search categories by name	Admin uses search box to filter categories	Admin is logged in	1. Enter text in search box. 2. Press Enter.	1. List filters to matching categories.
TC_UI_CAT_ADMIN_10	Admin can filter by parent category	Admin filters categories by parent dropdown	Admin is logged in	1. Select parent from filter dropdown. 2. Apply filter.	1. List shows only sub-categories of selected parent.
TC_UI_CAT_ADMIN_11	Admin can sort categories	Admin sorts category list by ID, Name, Parent	Admin is logged in	1. Click column headers for ID, Name, Parent. 2. Observe sorting.	1. List sorts accordingly. 2. Sort order toggles on second click.
TC_UI_CAT_ADMIN_12	Admin sees "No category found" message	Admin searches for non-existent category	Admin is logged in, no matching categories	1. Search for non-existent category. 2. Verify message.	1. "No category found" displayed.
TC_UI_CAT_ADMIN_13	Admin category form cancel button works	Admin cancels the add/edit category form	Admin is logged in	1. Click "Add Category". 2. Click Cancel.	1. Redirected to /ui/categories. 2. Form not submitted.
TC_UI_CAT_ADMIN_14	Admin category list updates dynamically	Admin sees list update after add/edit/delete without refresh	Admin is logged in	1. Perform add, edit, delete actions. 2. Verify list updates without refresh.	1. List updates immediately. 2. No page reload required.
UI Test Cases – User Role					
TC_UI_CAT_USER_01	User sees Category List (read-only)	User views category list without edit/delete options	User is logged in	1. Navigate to /ui/categories. 2. Verify list is displayed.	1. Category list visible. 2. No edit/delete buttons.
TC_UI_CAT_USER_02	User does not see "Add Category" button	User views category list without Add button	User is logged in	1. Navigate to /ui/categories. 2. Look for "Add Category" button.	1. "Add Category" button is not visible.
TC_UI_CAT_USER_03	User can search categories	User searches categories using search box	User is logged in	1. Enter text in search box. 2. Press Enter.	1. List filters accordingly.
TC_UI_CAT_USER_04	User can filter categories by parent	User filters categories using parent dropdown	User is logged in	1. Select parent from filter dropdown. 2. Apply filter.	1. Only sub-categories of selected parent shown.
TC_UI_CAT_USER_05	User can sort categories	User sorts category list by columns	User is logged in	1. Click column headers to sort. 2. Verify sorting.	1. List sorts correctly. 2. Sort order toggles.
TC_UI_CAT_USER_06	User cannot click edit/delete buttons	User views category list with disabled edit/delete options	User is logged in	1. Navigate to /ui/categories. 2. Look for edit/delete icons.	1. Edit/delete icons are not visible or disabled.
TC_UI_CAT_USER_07	User sees "No category found" for empty search	User searches for non-existent category	User is logged in	1. Search for non-existent category. 2. Verify message.	1. "No category found" displayed.
TC_UI_CAT_USER_08	User can navigate to category details	User clicks on category name to view details	User is logged in	1. Click on a category name. 2. Verify navigation.	1. Category details page loads. 2. URL reflects category ID.

TC_UI_CAT_USER_09	User cannot access /ui/categories/add	User attempts to access Add Category page directly	User is logged in	1. Navigate to /ui/categories/add directly. 2. Verify response.	1. Redirected to 403 page or denied access. 2. Error message displayed.